

box that contains the items to be delivered. The customer receives a password to open the locker box. Contactless authentication options may also be used.

Autonomous last-mile delivery robots vary in size and function. Some, like Nuro, are as big as a small car, while others are small and cute—like a locker box with wheels. Some move on pavements, and others on roads. Some can do only a single delivery, while others have modular containers and can manage multiple stops.

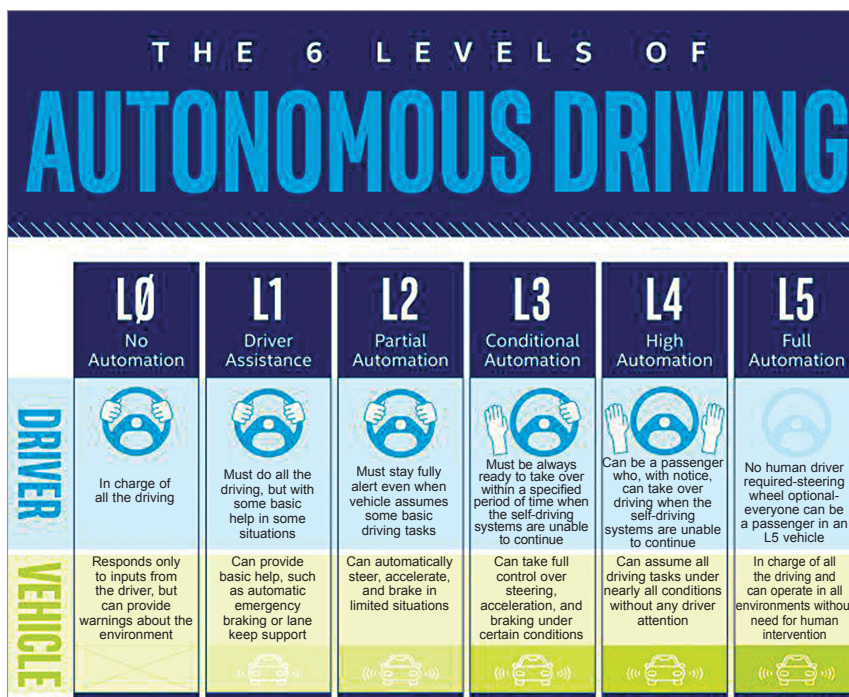
Amazon was one of the earliest to enter this field. They trialed a six-wheeled delivery device called Scout in California for almost three years, but they scaled back the project last year.

FedEx had also developed an interesting same-day delivery robot, Roxo, in collaboration with famous inventor Dean Kamen. A handsome robot, with flashy lights and all the works, it could even climb curbs and steps to delivery packages directly to a customer's door, making it more accessible to customers with limited mobility. However, they shut down the project last year to focus on more near-term opportunities.

Nevertheless, these early movers served as inspiration for a dozen or more startups to make this a very competitive space today!

Nuro was the first company to receive an autonomous exemption from the National Highway Traffic Safety Administration, USA. Its four-wheeled machine weighs nearly 700kg and has a top speed of around 70kmph. It can carry payloads of 225kg, with heating and cooling capabilities and customisable compartments. Nuro has piloted solutions in Texas, Arizona, and California for partners like Dominos, Kroger, Walmart, FedEx, and 7-Eleven. They recently signed a 10-year deal with Uber Eats for robot food delivery.

Not all robots are this big. Those



The six levels of autonomy in vehicles (Courtesy: Intel)

from Starship, Kiwibot, Cyan Robotics, and Ottonomy.IO are small and cute. Kiwibot has deployed over 50 robots in more than 25 universities, and has made more than 200,000 deliveries in the cities of San Jose, Santa Monica, Denver, and Dallas in the US, and in parts of Taiwan and Columbia.

Coco (Cyan Robotics) robots specialise in 15-minute food deliveries from local restaurants. Tortoise, on the other hand, debuted mobile vending robots. A vendor can load the Tortoise robot with goods. The robot goes around campuses or street sidewalks 'selling' these. Customers can take what they want and self-checkout. The robots are supervised remotely by human operators.

Ottobot from Ottonomy.IO makes life convenient for passengers at airports by delivering food near their respective gates. They operate in several airports, including Cincinnati, Pittsburgh, and Rome. At this year's CES, Ottonomy.IO revealed Yeti, a self-dispensing robot that can deliver goods even when the

customer is not available at home to collect it. It would be risky to deliver fragile objects though, because they slide down from the locker box, and there is a chance of breakage.

Goggo Network is an autonomous mobility solutions provider in Europe. Like most others, they also offer their solutions as-a-service to multiple partners. They have small and big, single- and multiple-stop robots. Last year, they joined the French government funded 5G Open Road project. They operate in France and Spain, with clients like Carrefour and DIA.

Swiss startup LOXO plans to launch its mobile delivery robots in public roads this spring. They plan to start with Switzerland and expand to the UK and Germany. Their AV is essentially a compact, electrically-powered box on wheels with compartments for packages that uses an array of sensors to deliver its self-driving functionality. LOXO travels at typical urban traffic speeds and stops immediately at the first sign of any danger. It is observed